How to AI an AI for Almost Anything

Lecture 5 – Multi-Agent Evolution

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Announcements

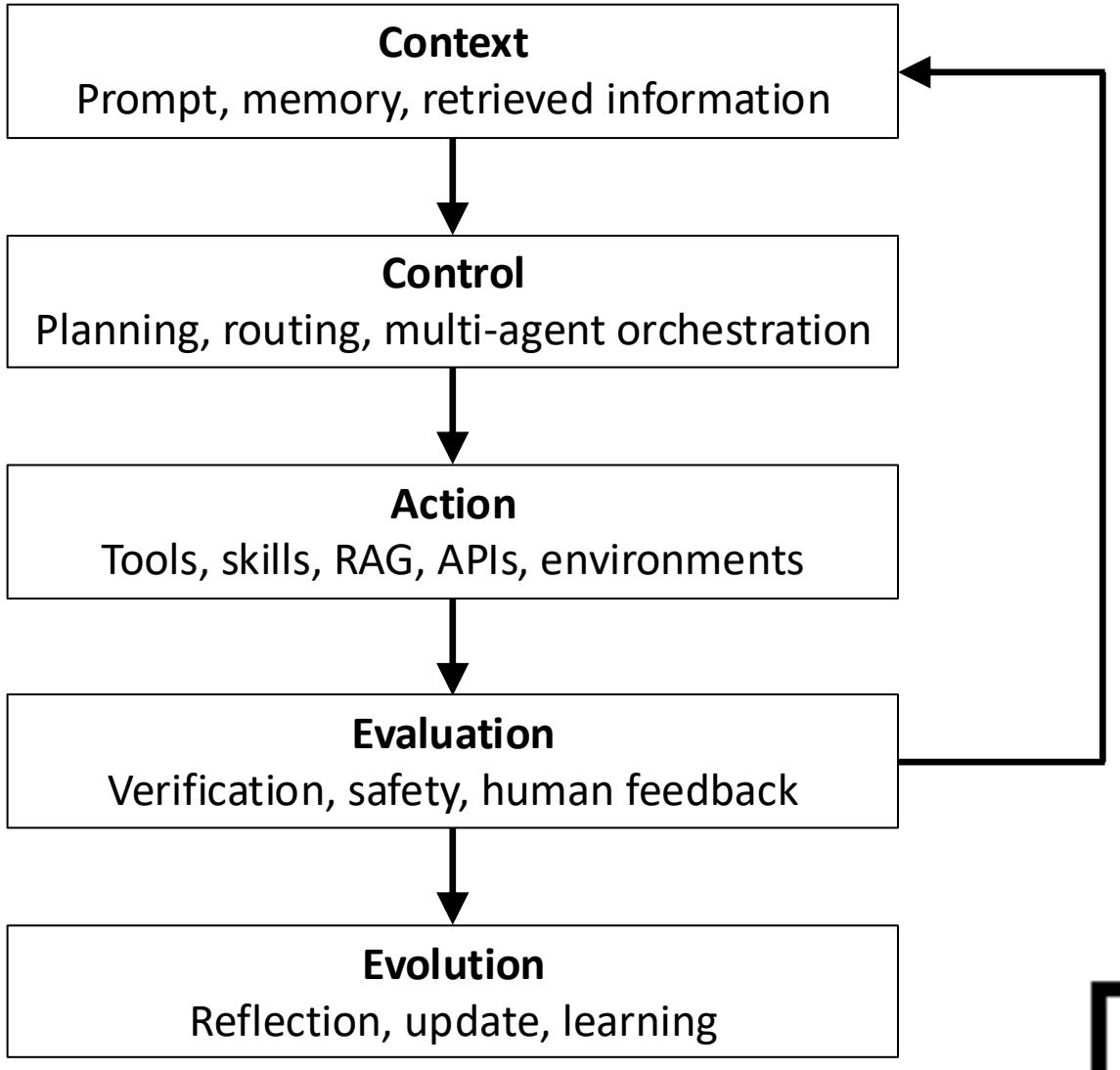
- Homework 1 extended, due Oct 15 11:59pm
- Work on projects, next checkpoint is midterm report + presentation Oct 28
  - Instructions posted on canvas
  - Processed datasets, ran baselines, done error analysis, better proposed idea
  - Bottom–up research. At the same time, continue top-down ideation from readings and literature review.

Today's Lecture

- Planning and task decomposition
- Multi-agent systems
- Multi-agent evolution

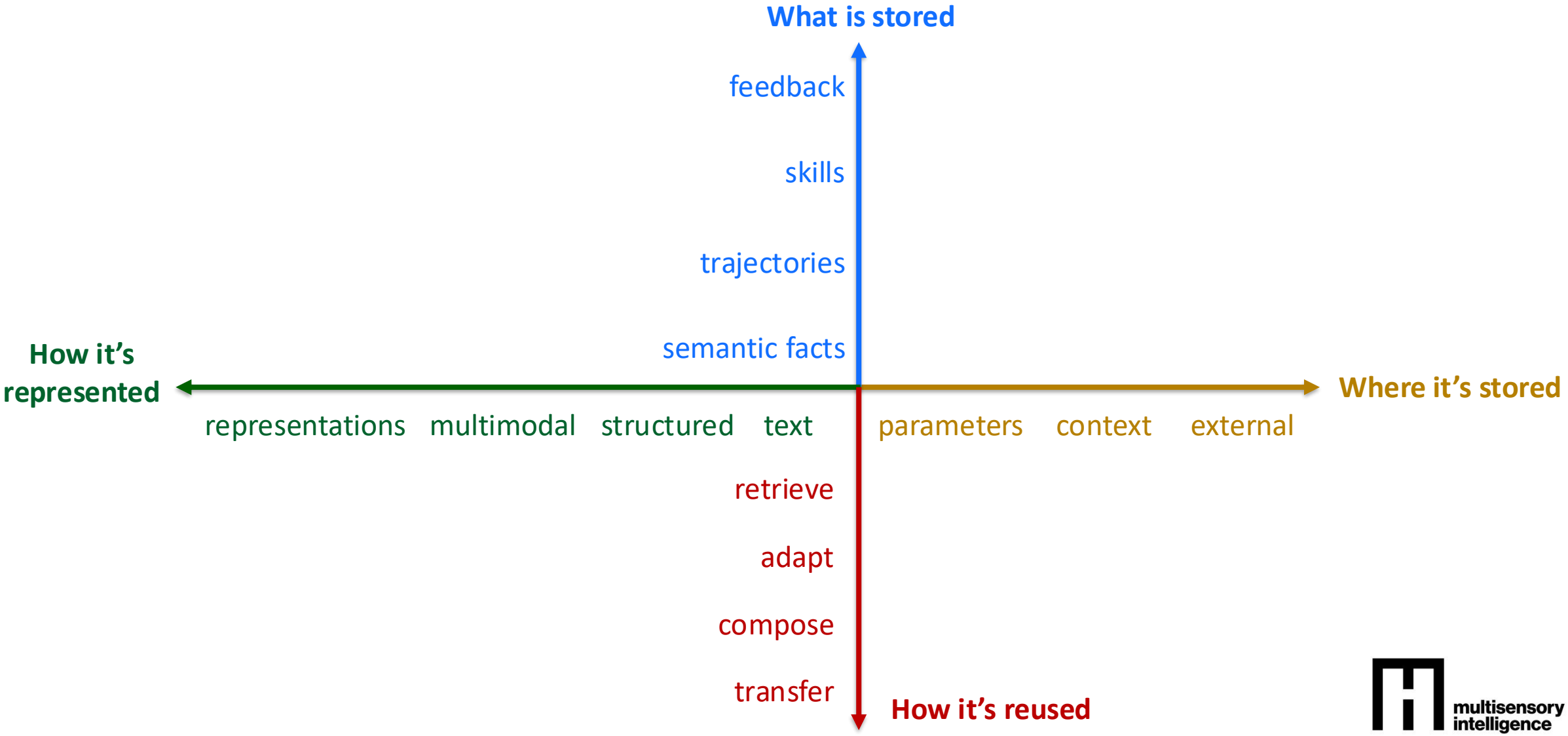
Recap of Harnesses

Harness: Code around the model that manages its context and execution flow to complete tasks



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Recap of Memory



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Recap: Harness Evolution

MEM1: treat memory as internal state in context autonomously updated by LLM

CLM: treat entire context as persistent file autonomously maintained and rewritten by LLM

Meta-Harness: treat entire harness as a file system autonomously modified by LLM

What's common about all these methods?

- Simple, no hand engineering, end-to-end
- New paradigm that generalizes what can be autonomously optimized (memory -> context -> harness)
- Scalable results for zero-shot, in-context prompting, fine-tuning

RLM: more structure in task decomposition and context management with recursive calls to sub-agents. but still highly scalable!

Planning and Task Decomposition

TASK

Buy the lab's new GPU workstation: compare configurations and prices across vendor sites, check compatibility against the cluster requirements on the internal wiki, and file the purchase requisition.

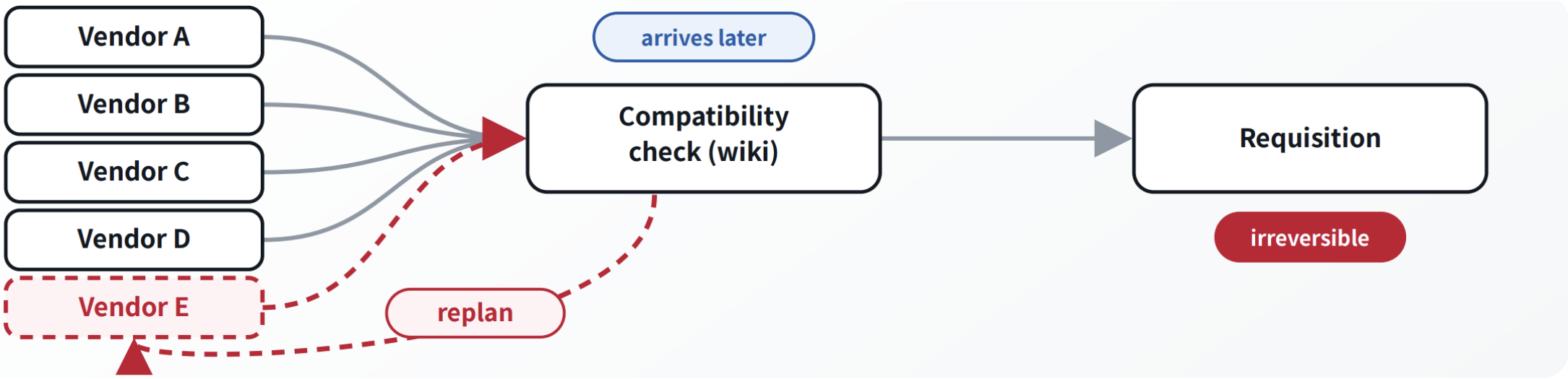
```
 1 ACTION   goto vendor-a.com → Workstations → spec sheet → GPU
            comparison ...
15 ACTION   configure: 2× RTX 6000, 128 GB RAM, 850 W PSU
31 OBSERVE  wiki: “racks accept 4U chassis; each slot with 1600 W”
        context compacted · 128k → 14k tokens.
    four vendors searched: only D fits, but has 20-week lead time
201 ACTION  pick D? look for a fifth vendor? relax a requirement?
202 ACTION  goto procurement portal → New requisition → Submit
```

1. **Four independent searches run as one thread in one context.**
2. **The four sites it chose all come back and none is great.**
3. **It picks one anyway and files the requisition.**

[credit: CMU AI Agents course fall 2026: https://www.cmu-agents.com/]

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Planning and Task Decomposition



The task has parallel searches, dependencies, information that arrives later in the task, and a step that cannot be undone.

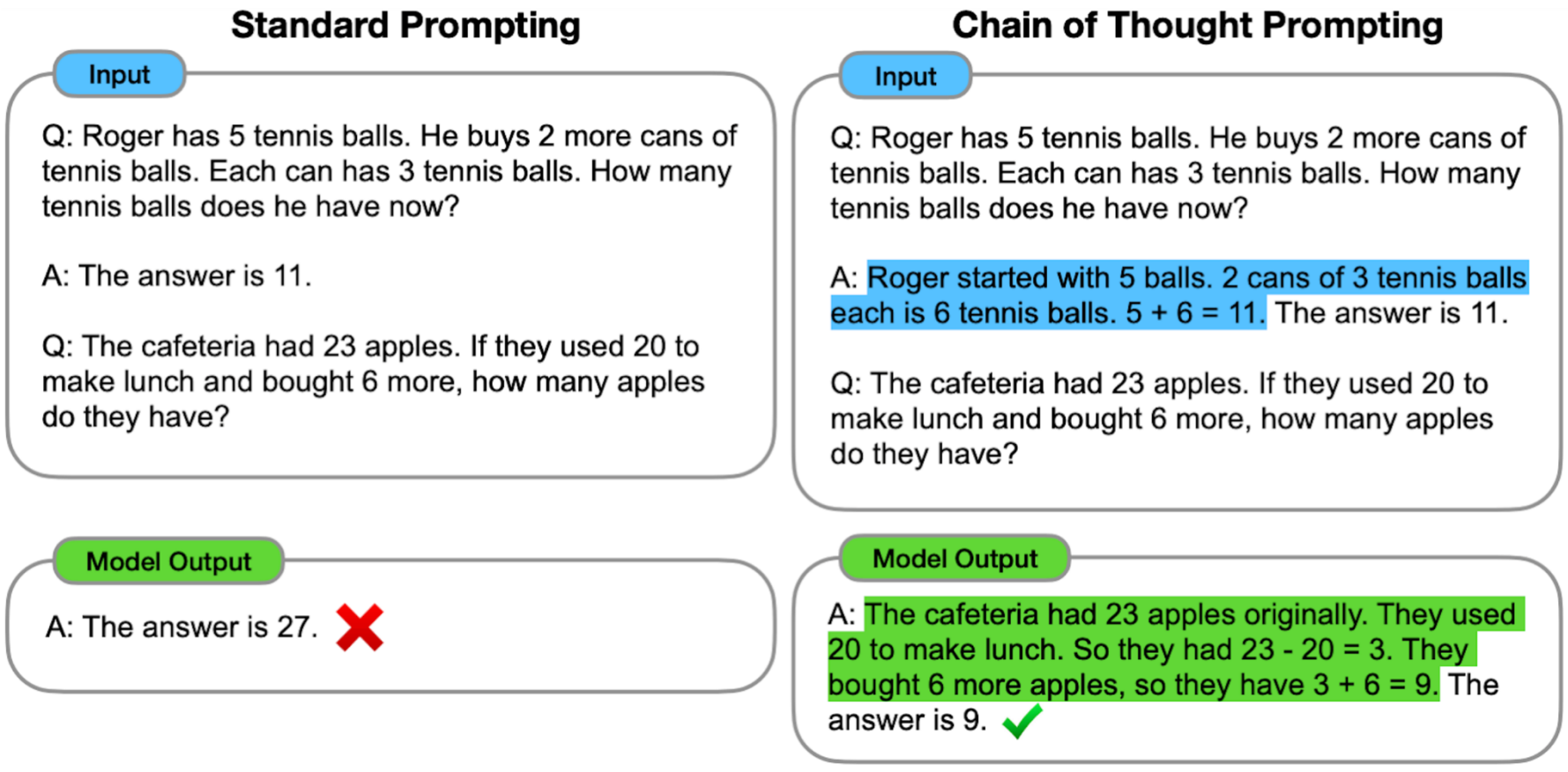
The graph itself also changes: when none of the four vendors fits, a fifth search has to be added.

Key questions:

1. When is planning useful and when can it hurt?
2. How to represent plans?
3. When is a plan made and when should it change?

[credit: CMU AI Agents course fall 2026: https://www.cmu-agents.com/]

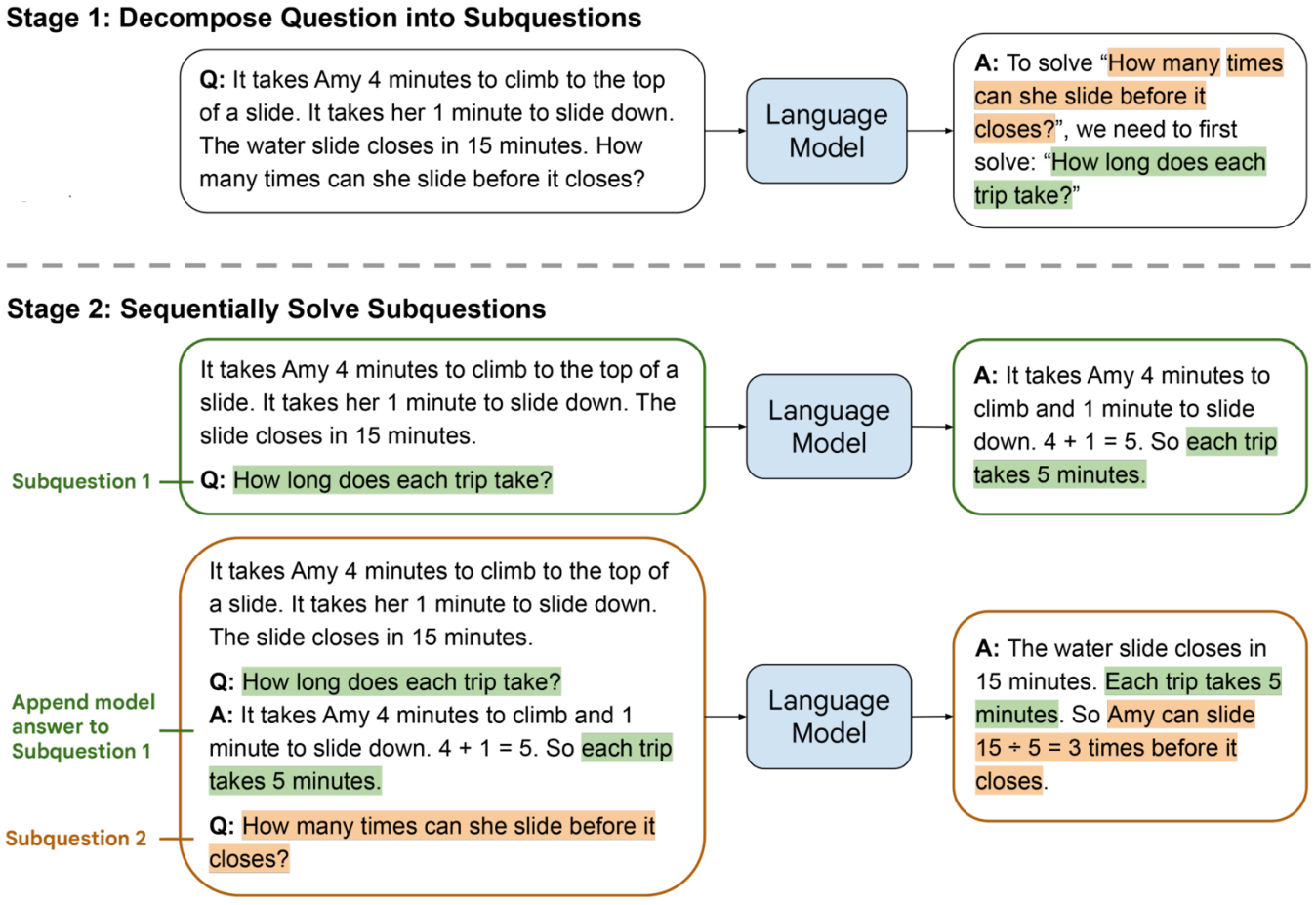
Planning via Prompting



[Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022]

Planning via Prompting

Enables modularity, specialized models, separates planner from executor.



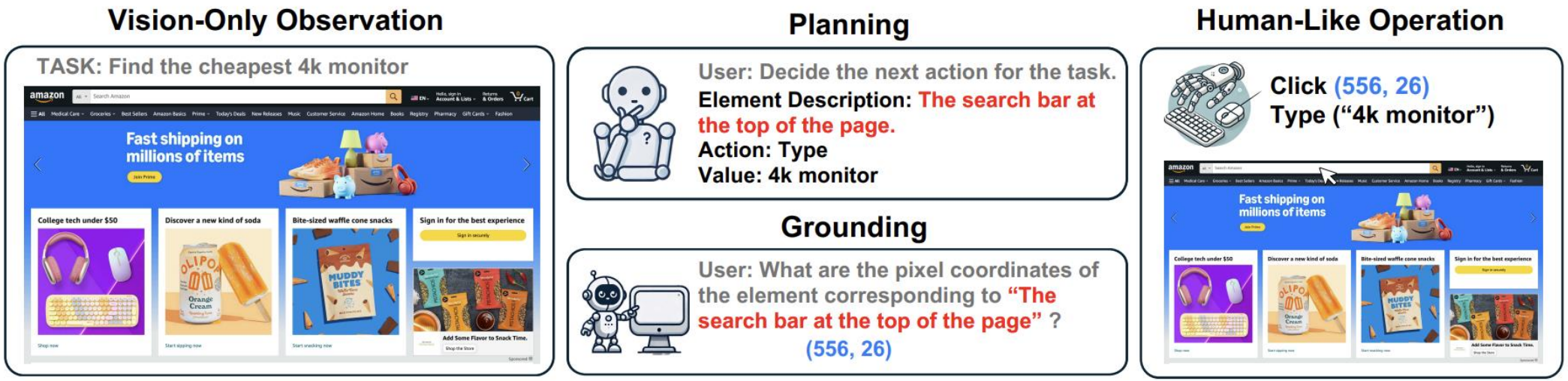
Prompt-based planning works in text-only, action-free, reversible scenarios.

If plans act on the world, changes the state of the agent, or cannot be reversed, then need to plan and re-plan.

[Zhou et al., Least-to-Most Prompting Enables Complex Reasoning in LLMs. ICLR 2023]

Guidance and Orchestration Models

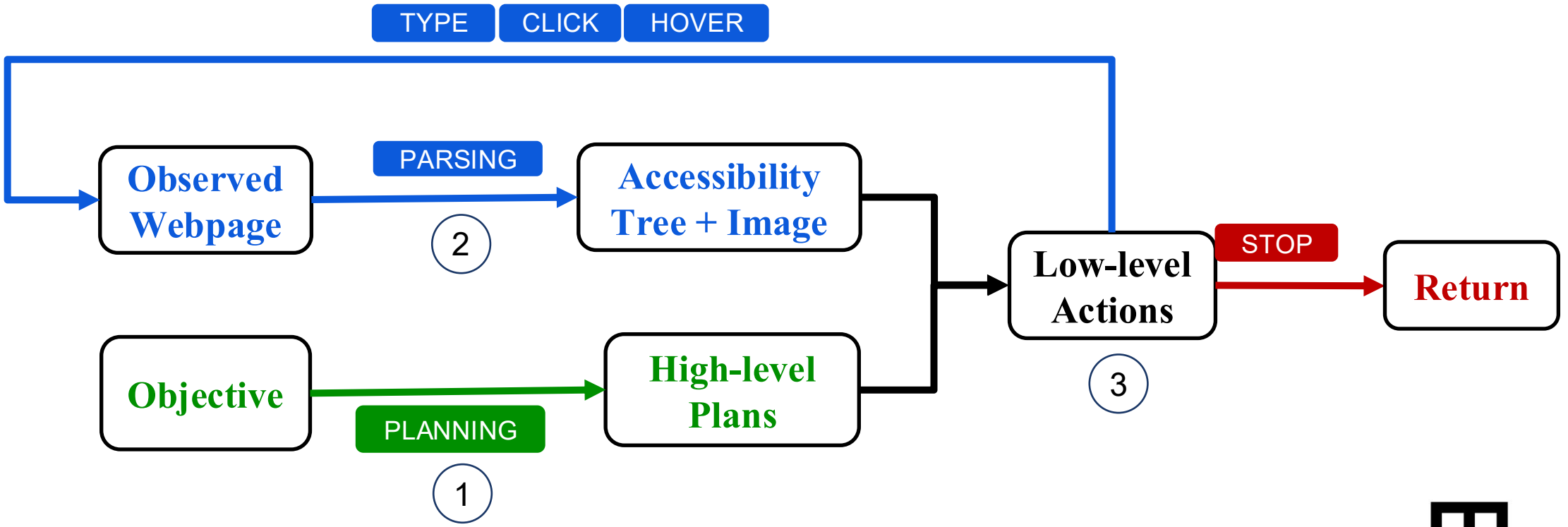
Planning model is larger and visually grounded, but bad at producing low-level actions. Separate small 7B execution model maps plans to coordinate actions.



[Gou et al., Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents. ICLR 2025]

Planning in Agents

- General architecture of web agents:
  - High-level Reasoning
  - Observation Parsing
  - Low-level Action Generation



[Zhou et al., WebArena: A Realistic Web Environment for Building Autonomous Agents. ICLR 2024]
[Jang et al., VideoWebArena: Evaluating Multimodal Agents on Video Understanding Web Tasks. ICLR 2025]

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Planning with Human-in-the-loop

Asking humans for clarification when it is uncertain about generated plans

What is the price range of wireless earphone in the One Stop Market?

Zero-shot high-level plan

[Search] wireless earphone in the One Stop Market

[Find] the price range of wireless earphone in the One Stop Market

Too ambiguous

Few-shot high-level plan

[Search] wireless earphone in the One Stop Market

[Find] the price of the first wireless earphone

[Read] the price range of wireless earphones in the One Stop Market

Didn't sort

Few-shot high-level plan with human clarification

[Search] wireless earphone in the One Stop Market

[Sort] the price of the wireless earphones from low to high

[find] the lowest priced wireless earphone

[sort] the price of wireless earphones from high to low

[find] the highest priced wireless earphone

Subsequent model generations become correct after clarification

Ground truth high-level plan

[Search] wireless earphones in the One Stop Market.

[Sort] the price of the wireless earphones from low to high

[Find] the first item

[Sort] the price of the wireless earphones from high to low

[Find] the first item

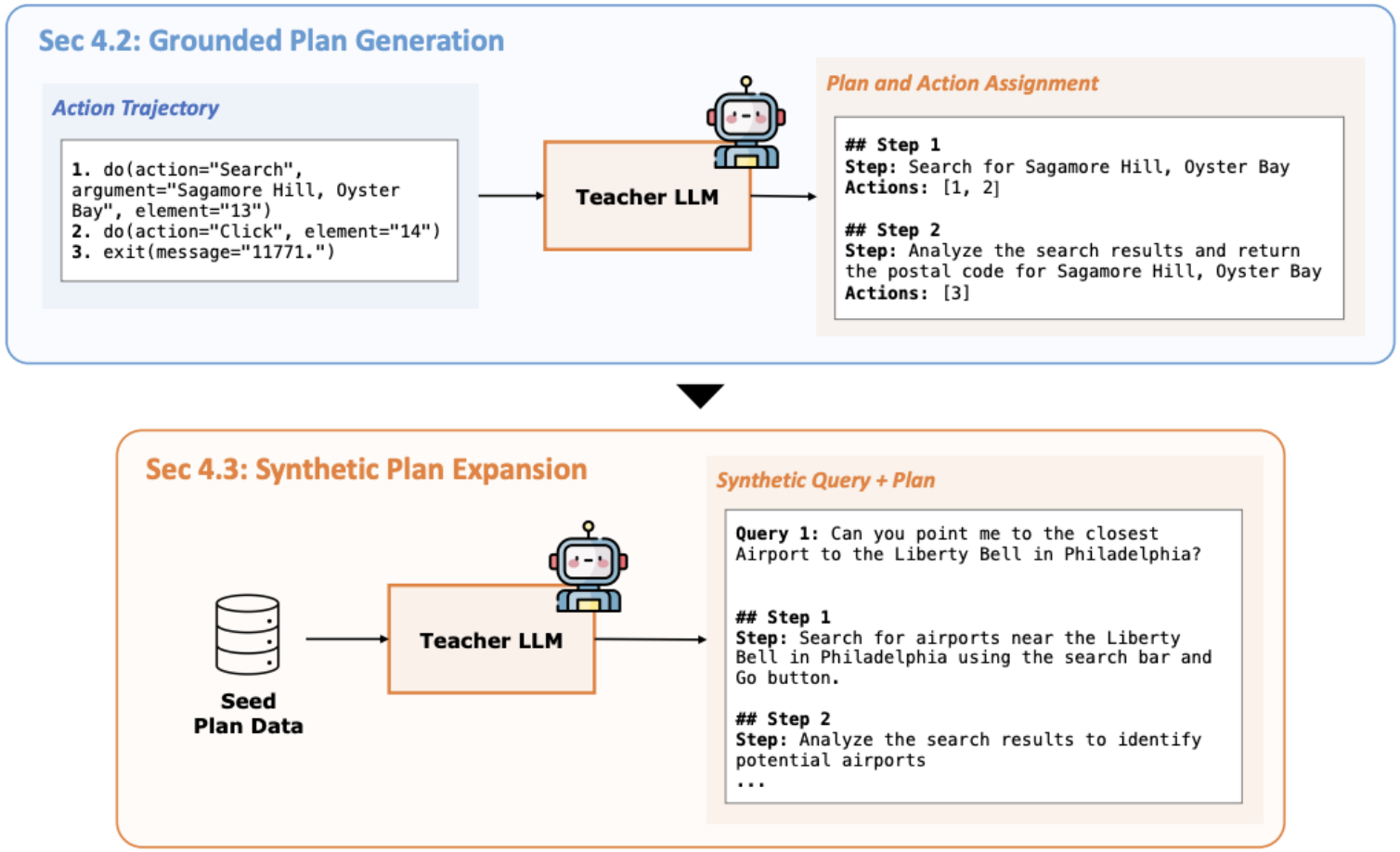
[Identify] the price range of wireless earphones in the One Stop Market

[Kaufmann et al., A Survey of Reinfocement Learning from Human Feedback. arXiv 2024]

[Wang et al., Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels. CHI 2024]

Training the Planner

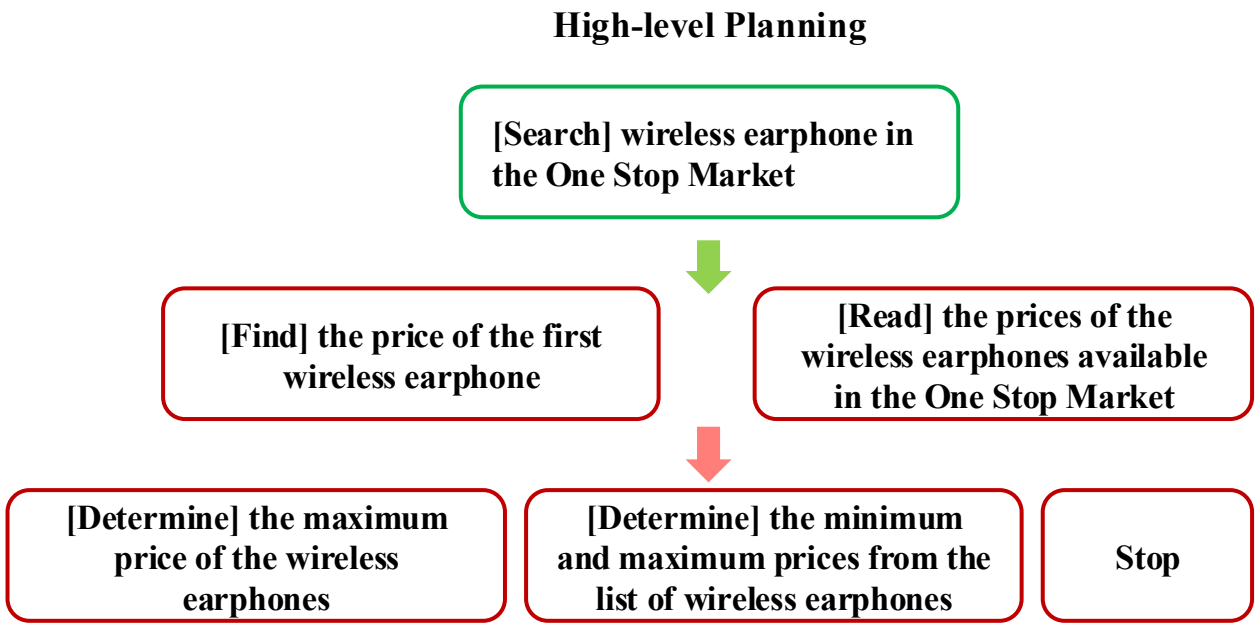
Synthetic plan generation by teacher model + supervised fine-tuning of planner model and execution model.

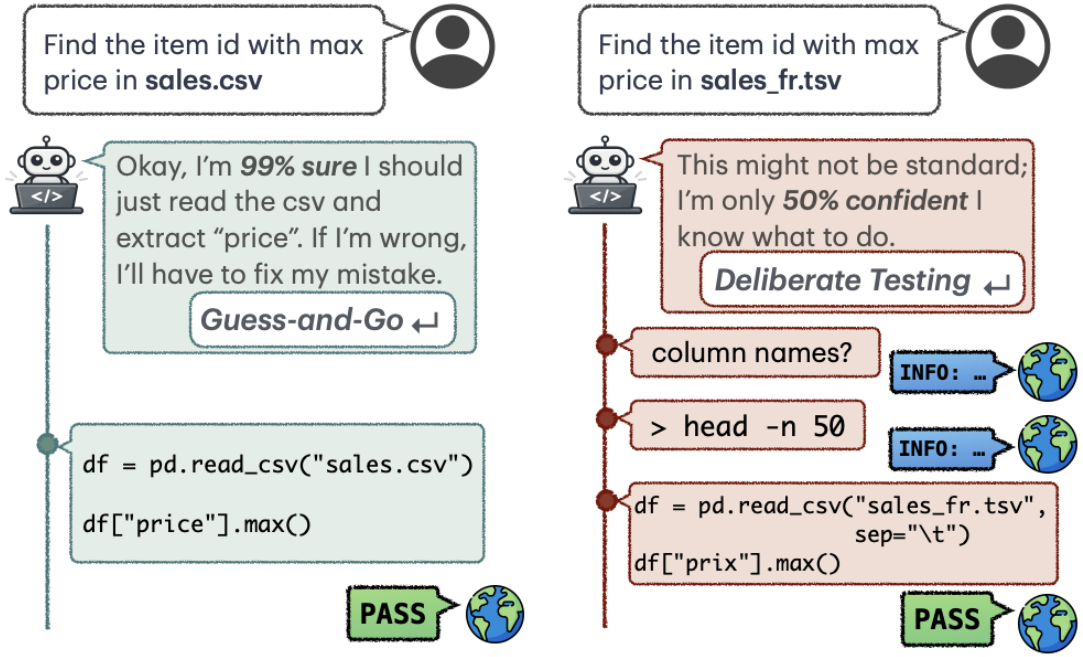


[Erdogan et al., Plan-and-Act: Improving Planning of Agents for Long-Horizon Tasks. ICML 2025]

Calibrated Planning

Sampling high-level plans generated by the model multiple times to estimate uncertainty





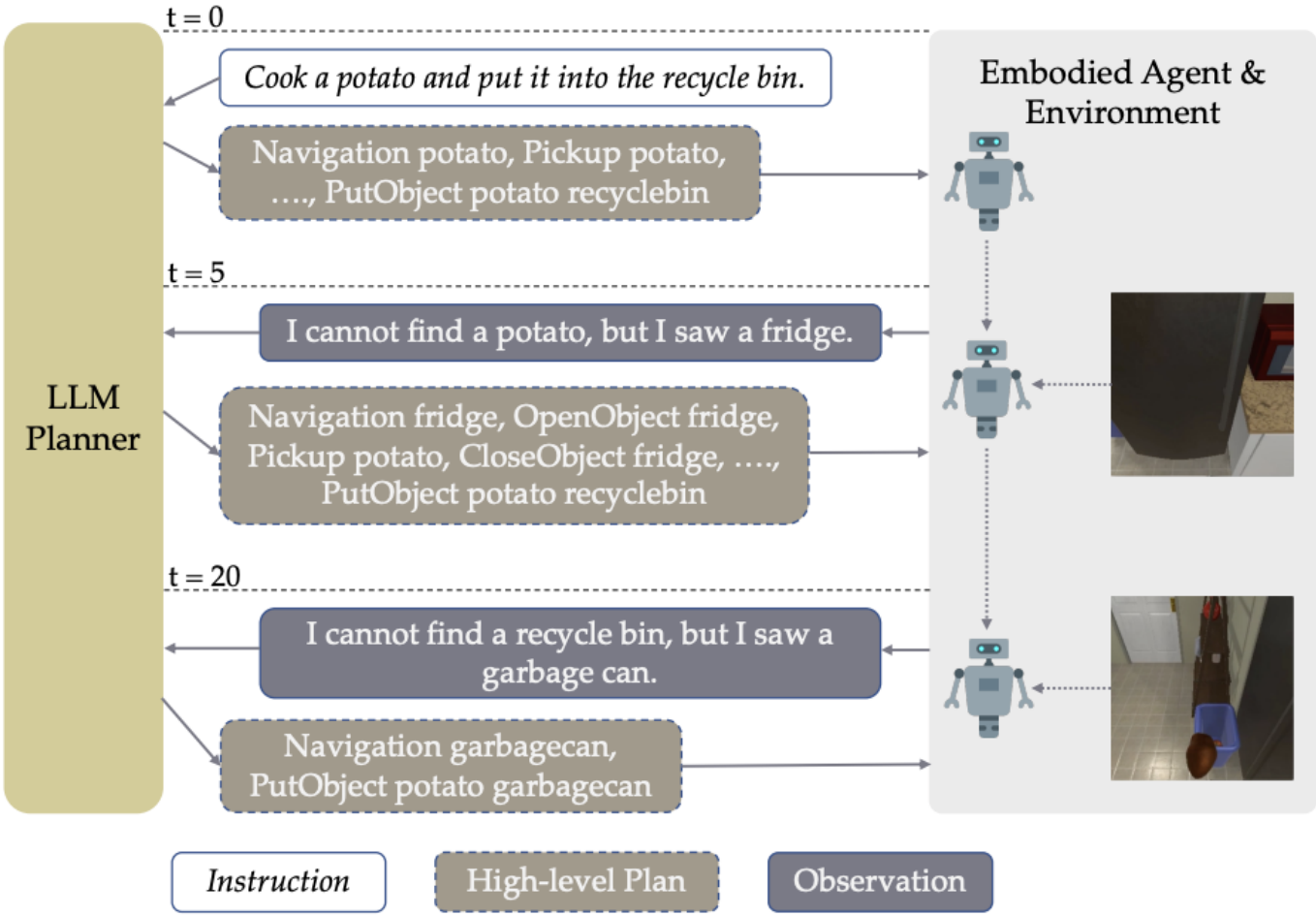
The optimal strategy is a function of model's ***success probability*** and ***environment costs*** of different actions. **Can LLMs behave optimally?**

[Ding et al., Calibrate-then-Act: Cost-Aware Exploration in LLM Agents. arXiv 2026]

Replanning

Planning, acting in embodied environment, observing feedback, plan again in response to new information or failures.

Main challenge: requires taking safe & non-reversible actions.



[Song et al., LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models. ICCV 2023]

Test-Time Planning and Interaction

Not just thinking, but actually executing the action, seeing feedback, and re-trying.

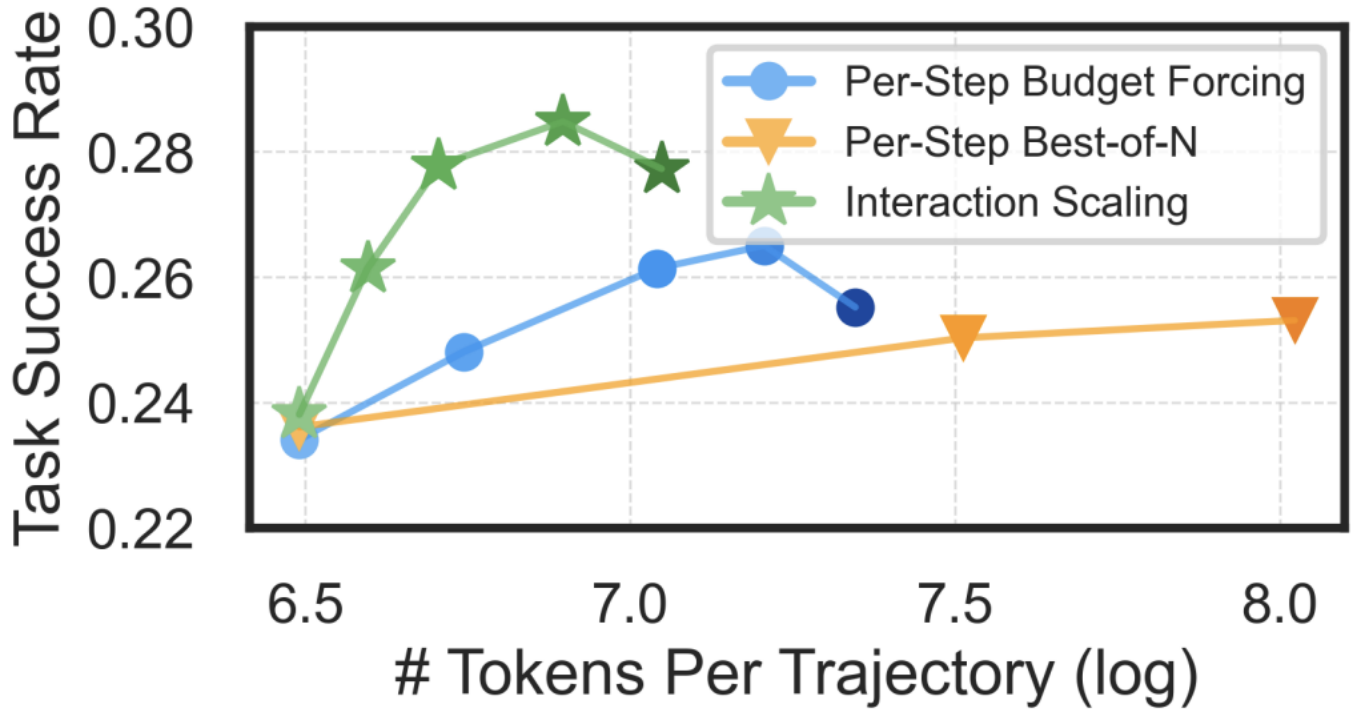


Figure 1: ***We propose a new-axis of test-time scaling for agents: scaling the number of interaction steps***. Unlike traditional methods that emphasize longer reasoning per step, we show that acting more helps gain new information from the environment and improve task performance (detailed results of the left plot in Section 4.2).

[Shen et al., Thinking vs Doing: Agents that Reason by Scaling Test-Time Interaction. NeurIPS 2025]

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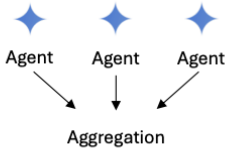
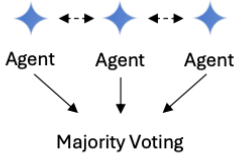
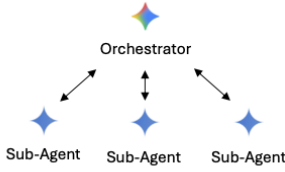
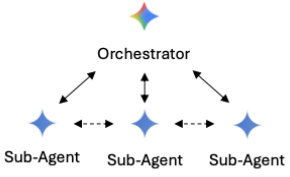
Putting Everything Together: Multi-Agent Systems

Planner/orchestration agents + execution agents.

Decentralized agents with aggregation function.

Multimodal agents each processing different modalities.

Table 2 | Architectural comparison of agent methods with objective complexity metrics. Computational complexity measured in terms of LLM calls, coordination overhead, and parallelization potential.

Characteristic	SAS	MAS (Independent)	MAS (Decentralized)	MAS (Centralized)	MAS (Hybrid)
Interaction Type					
LLM Calls	$O(k)$	$O(nk) + O(1)$	$O(dnk) + O(1)$	$O(rnk) + O(r)$	$O(rnk) + O(r) + O(p)$
Sequential Depth	k	k	d	r	r
Comm. Overhead	0	1	$d \cdot n$	$r \cdot n$	$r \cdot n + p \cdot m$
Parallelization Factor	1	n	n	n	n
Memory Complexity	$O(k)$	$O(n \cdot k)$	$O(d \cdot n \cdot k)$	$O(r \cdot n \cdot k)$	$O((r + p) \cdot n \cdot k)$
Coordination	Sequential	Parallel + Synthesis	Sequential Debate	Hierarchical	Hierarchical + Peer
Consensus	-	Synthesis	Debate	Orchestrator	Orchestrator

[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

Decentralized Multi-Agent Systems

Multiple instances of a language model propose answers, critique one another, and debate over several rounds before reaching a final answer.

Question: Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Round 1

ChatGPT: Before the restart, Carla was able to download 40% of the file, which is 0.4 * 200 GB = 80 GB. At a rate of 2 GB/minute, it would take her: time = 80 GB ÷ 2 GB/minute = 40 minutes.

Because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the remaining 120 GB is: time = 120 GB ÷ 2 GB/minute = 60 minutes.

But Carla also had to wait 20 minutes for the Windows update to install, so the total time it took to download the file is: total time = 40 minutes + 60 minutes + 20 minutes = **120 minutes**. ✗

Bard: Carla is downloading a 200 GB file at a speed of 2 GB/minute. This means that it will take her 200 / 2 = 100 minutes to download the file.

However, Windows forces a restart to install updates, which takes 20 minutes. This means that Carla will have to restart the download from the beginning. Since Carla has already downloaded 40% of the file, she only needs to download 200 - 40 = 160 GB. At a speed of 2 GB/minute, it will take Carla 160 / 2 = 80 minutes to download the file.

Adding the time for the Windows restart, it will take Carla 80 + 20 = **100 minutes** to download the file. ✗

Round 2

ChatGPT : Based on the calculations provided by the other agent, the correct answer is: Before the restart, Carla was able to download 40% of the file, which is 0.4 * 200 GB = 80 GB. At a rate of 2 GB/minute, it would take her: time = 80 GB ÷ 2 GB/minute = 40 minutes

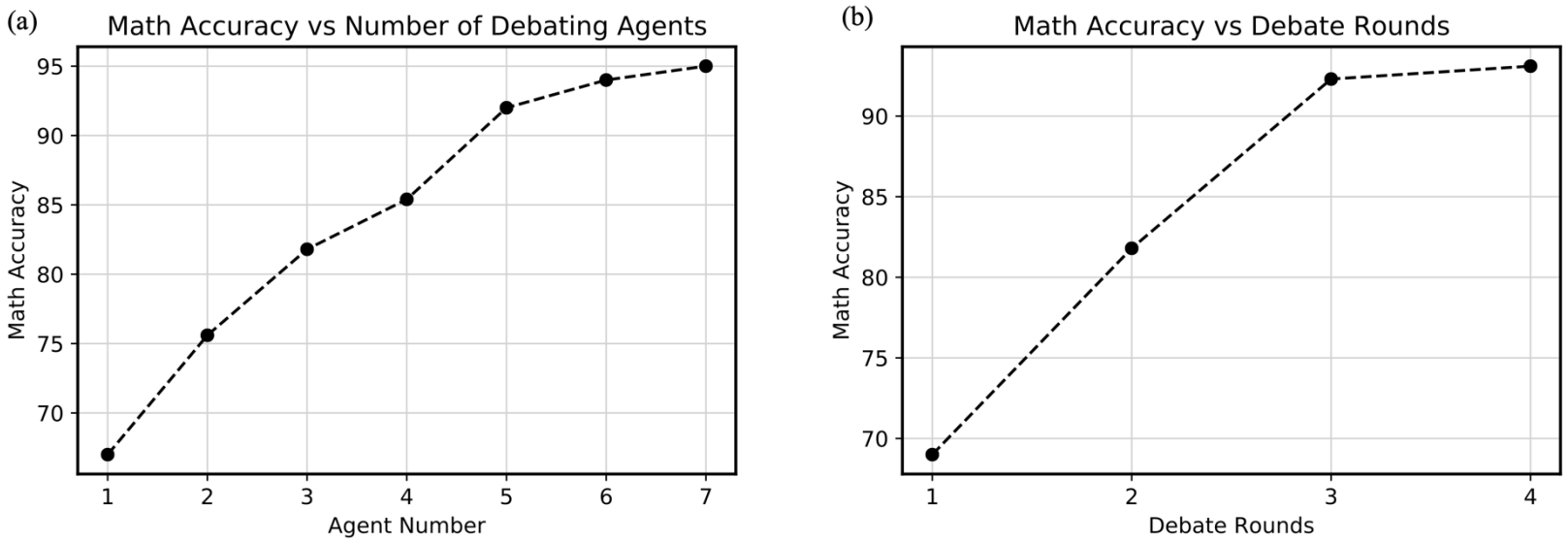
However, because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the entire 200 GB file is: time = 200 GB ÷ 2 GB/minute = 100 minutes

And since Carla had to wait 20 minutes for the Windows update to install, the total time it took to download the file is: total time = 40 minutes + 100 minutes + 20 minutes = **160 minutes**. ✓

[Du et al., Improving Factuality and Reasoning in LMs Through Multiagent Debate. ICML 2024]

Decentralized Multi-Agent Systems

Multiple instances of a language model propose answers, critique one another, and debate over several rounds before reaching a final answer.



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Centralized Multi-Agent Systems

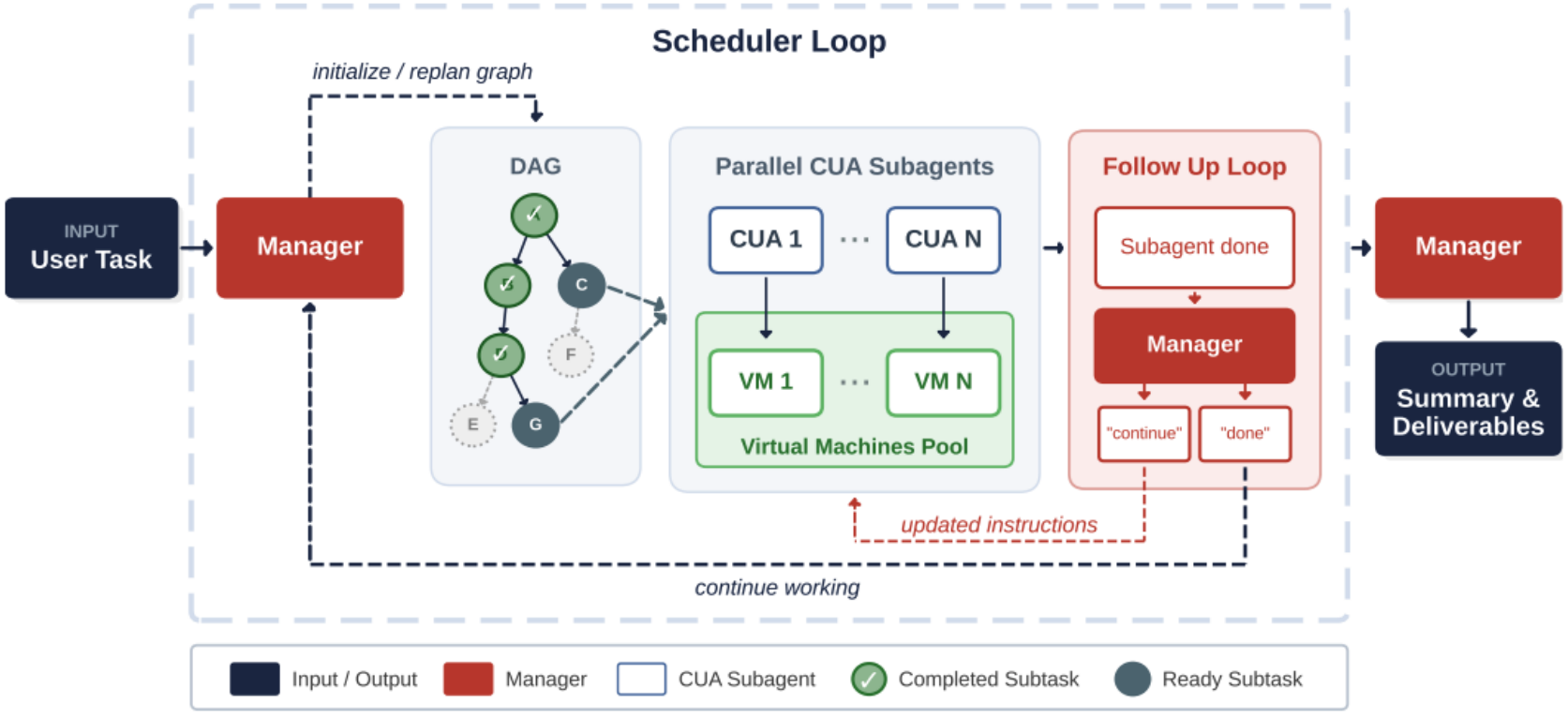
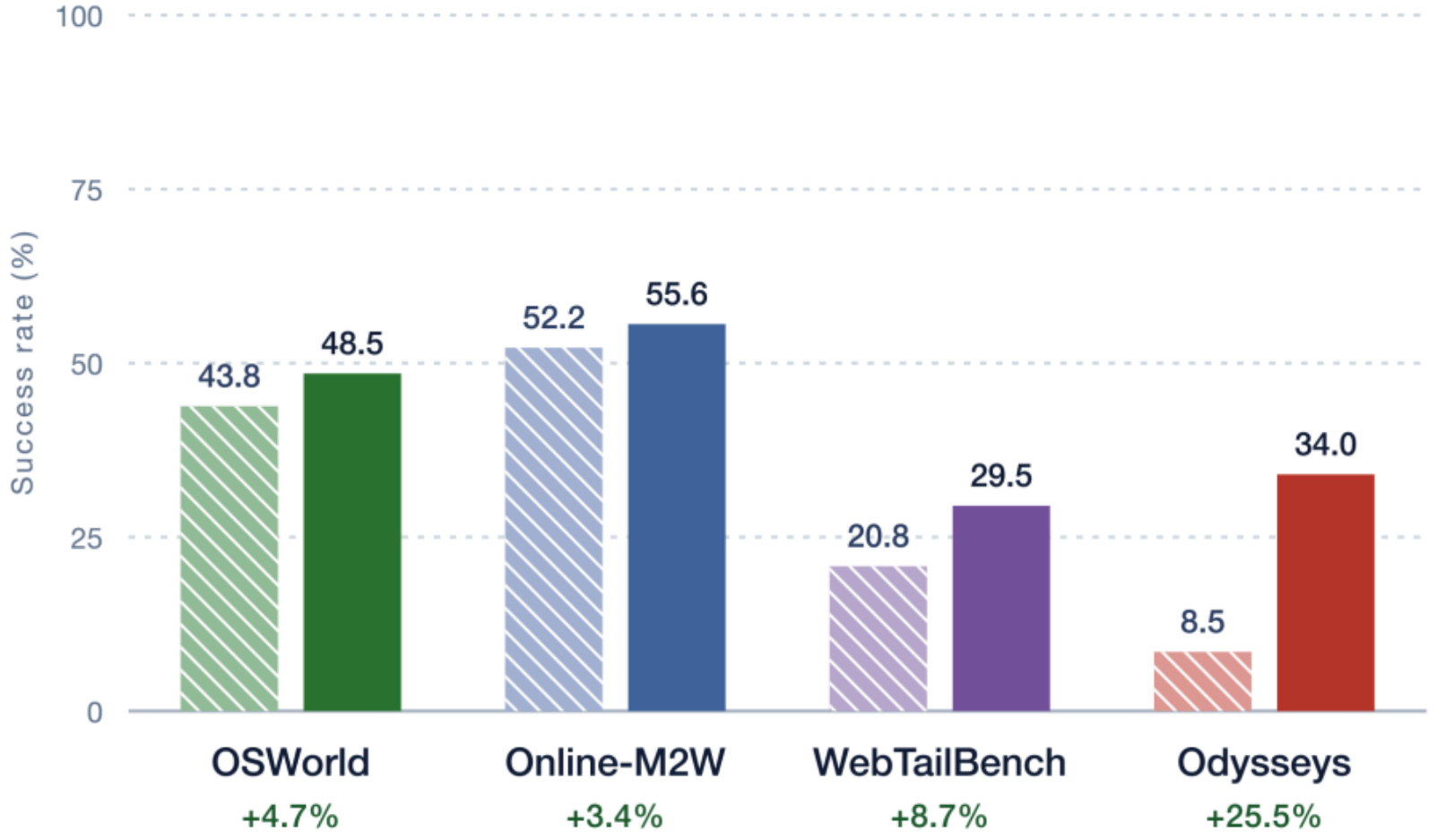


Figure 1: Overview of the setup for multi-agent computer use (MACU). At each iteration, ready subtasks (grey) are launched and handled in parallel by CUA subagents. The manager replans and updates the DAG based on information it receives from completed subagents.

[Koh et al., Multi-Agent Computer Use. COLM 2026]

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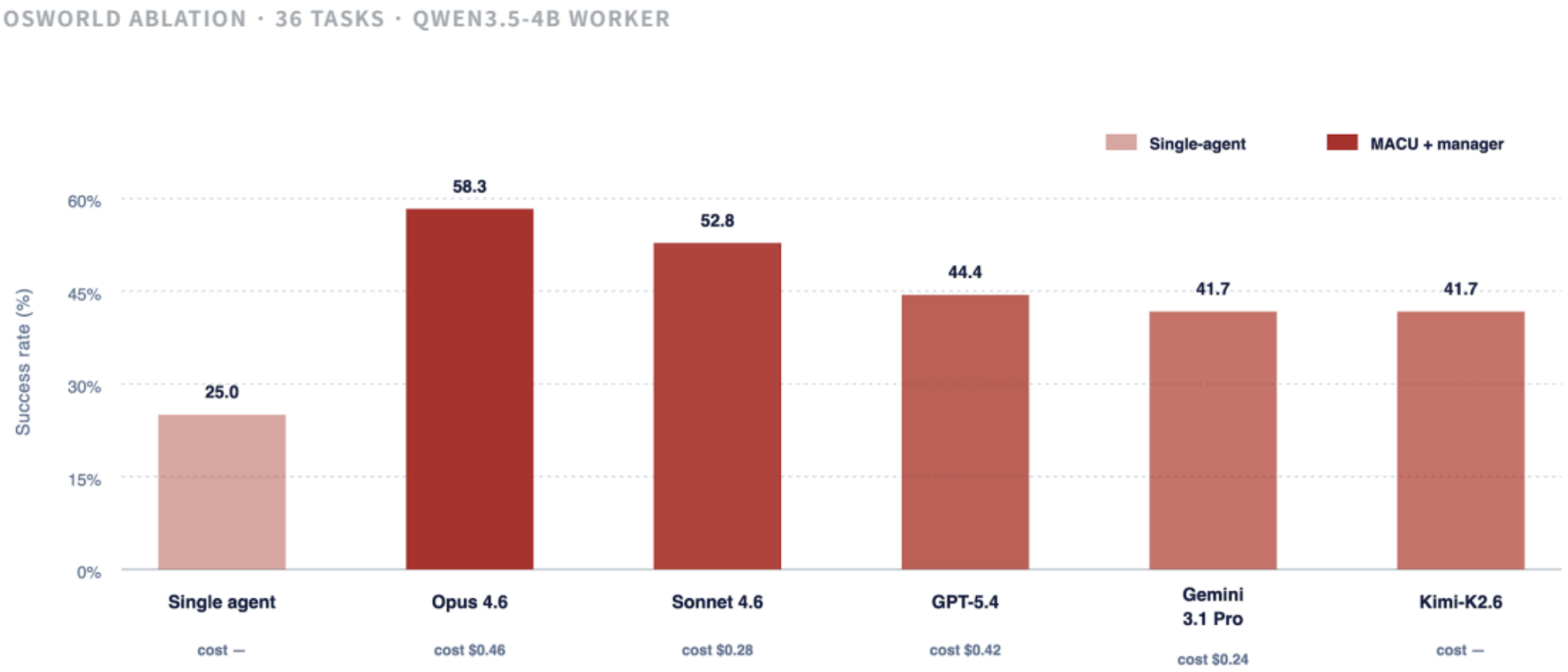
Centralized Multi-Agent Systems



[Koh et al., Multi-Agent Computer Use. COLM 2026]

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Importance of Strong Orchestrator



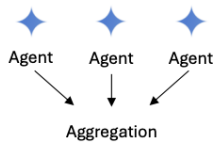
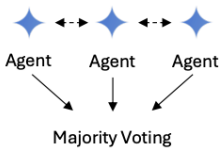
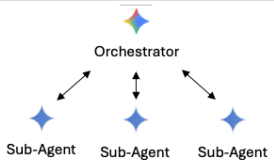
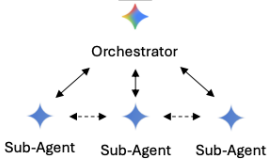
[Koh et al., Multi-Agent Computer Use. COLM 2026]

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When Does Multi-Agent Work?

More agents does not automatically improve performance: scaling depends on task structure and multi-agent topology.

Table 2 | Architectural comparison of agent methods with objective complexity metrics. Computational complexity measured in terms of LLM calls, coordination overhead, and parallelization potential.

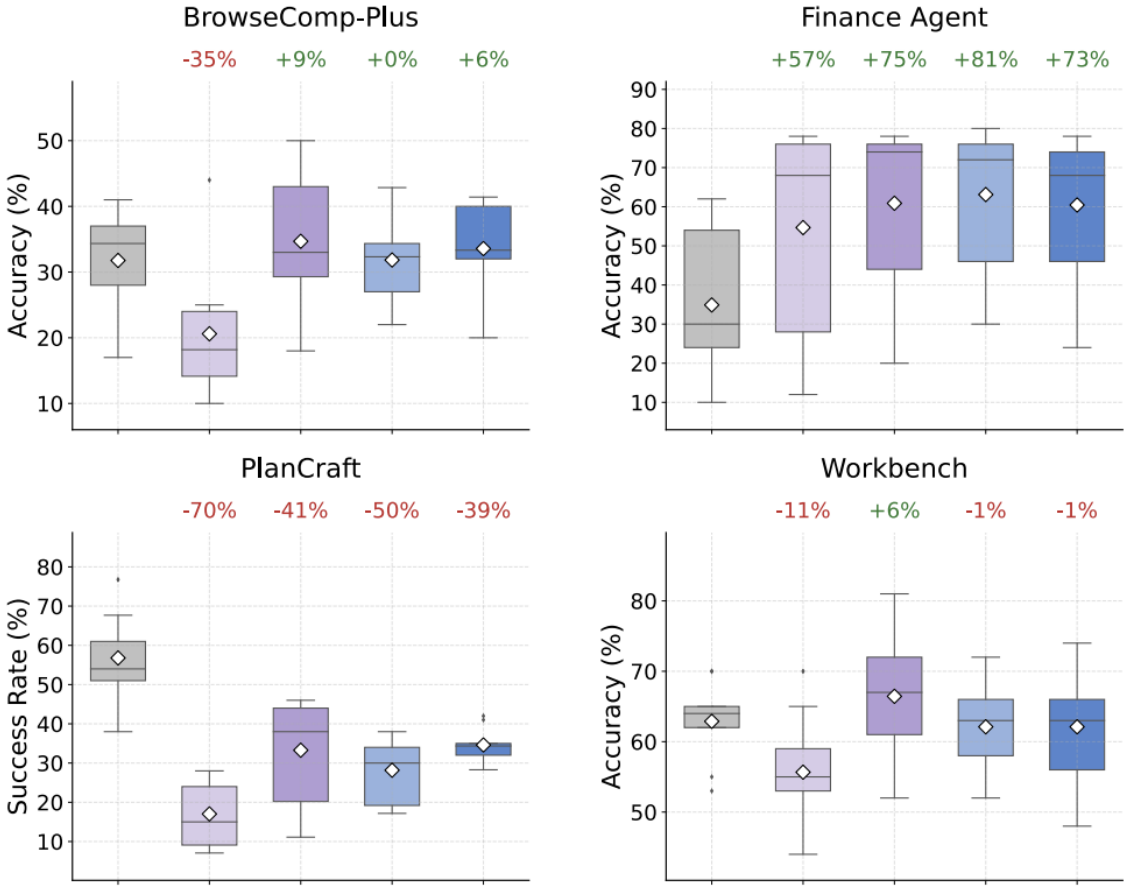
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Comm. Overhead	0	1	$d \cdot n$	$r \cdot n$	$r \cdot n + p \cdot m$
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Memory Complexity	$O(k)$	$O(n \cdot k)$	$O(d \cdot n \cdot k)$	$O(r \cdot n \cdot k)$	$O((r+p) \cdot n \cdot k)$
Coordination	Sequential	Parallel + Synthesis	Sequential Debate	Hierarchical Orchestrator	Hierarchical + Peer Orchestrator
Consensus	-	Synthesis	Debate	Orchestrator	Orchestrator

[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

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When Does Multi-Agent Work?

Finding 1: **Multi-agent systems only help for certain types of tasks.**



Ex 1. Finance agent (MAS wins +81%)
-> Task naturally decomposes (revenue/cost/market)

Ex 2. PlanCraft (MAS loses -70%)
-> Sequential dependency (action 1 affects action 2)
-> Splitting tasks hurts rather than helps

Guidelines:
Parallelizable tasks -> centralized
Exploration needed -> decentralized
Sequential dependency -> single-agent

[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

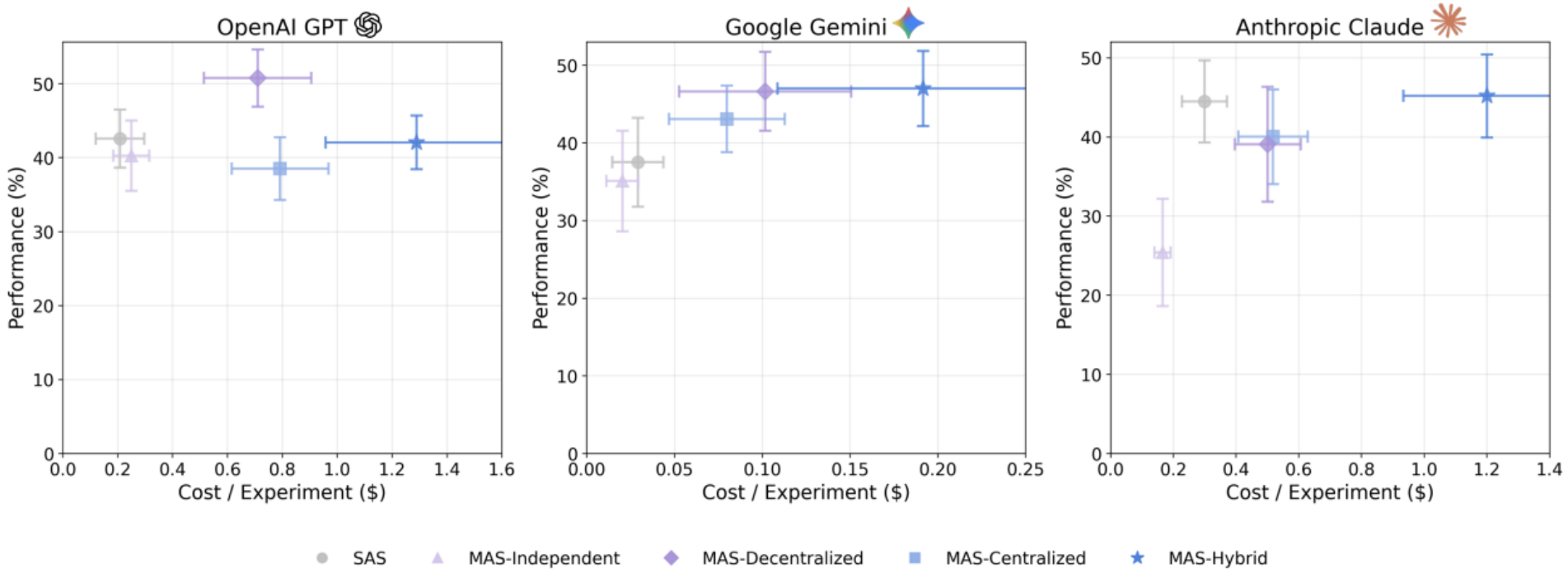
When Does Multi-Agent Work?

Finding 2: **Different base models exhibit different cost-performance tradeoffs.**

OpenAI: performance plateau despite increasing cost.

Google: higher cost, higher performance.

Anthropic: MAS can hurt performance.

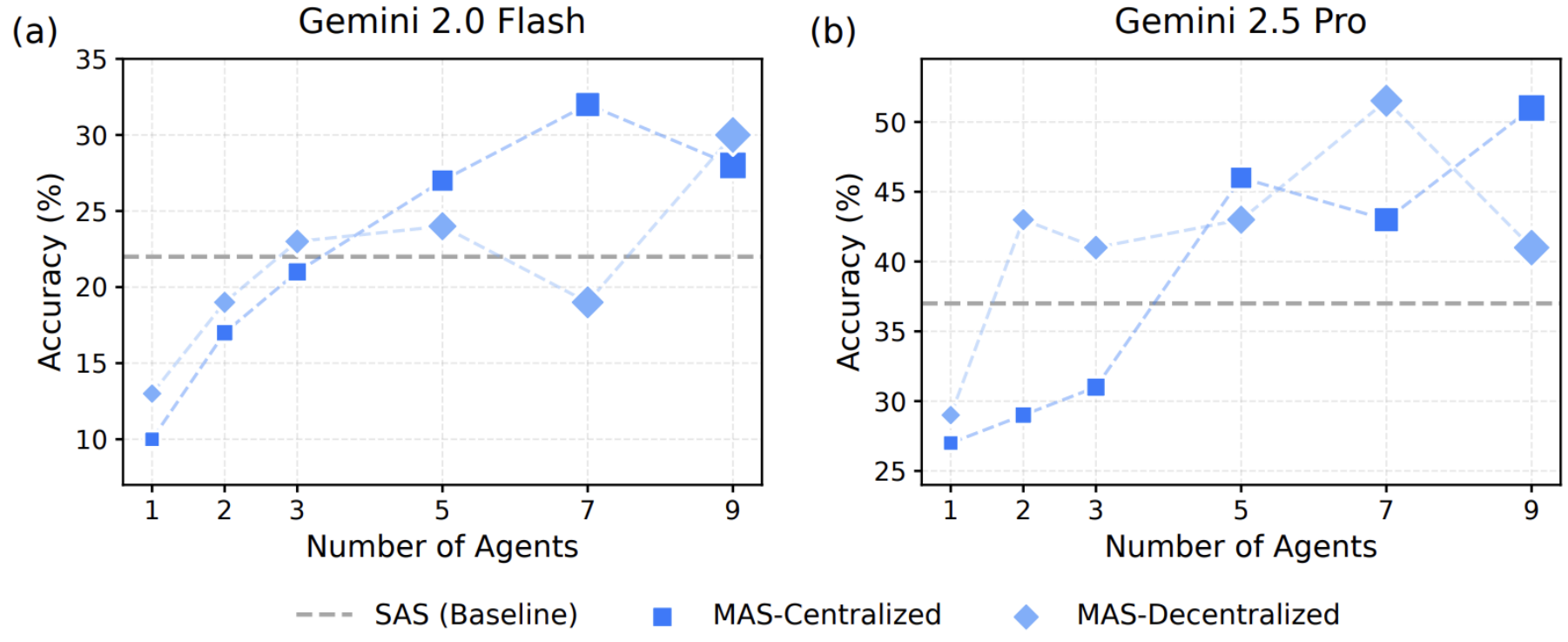


[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

When Does Multi-Agent Work?

Finding 3: **Increasing number of agents does not always improve performance.**

Tool-heavy tasks (e.g., 16-tool workflows) suffer from multi-agent coordination overhead, with penalties compounding as environment complexity increases.



[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

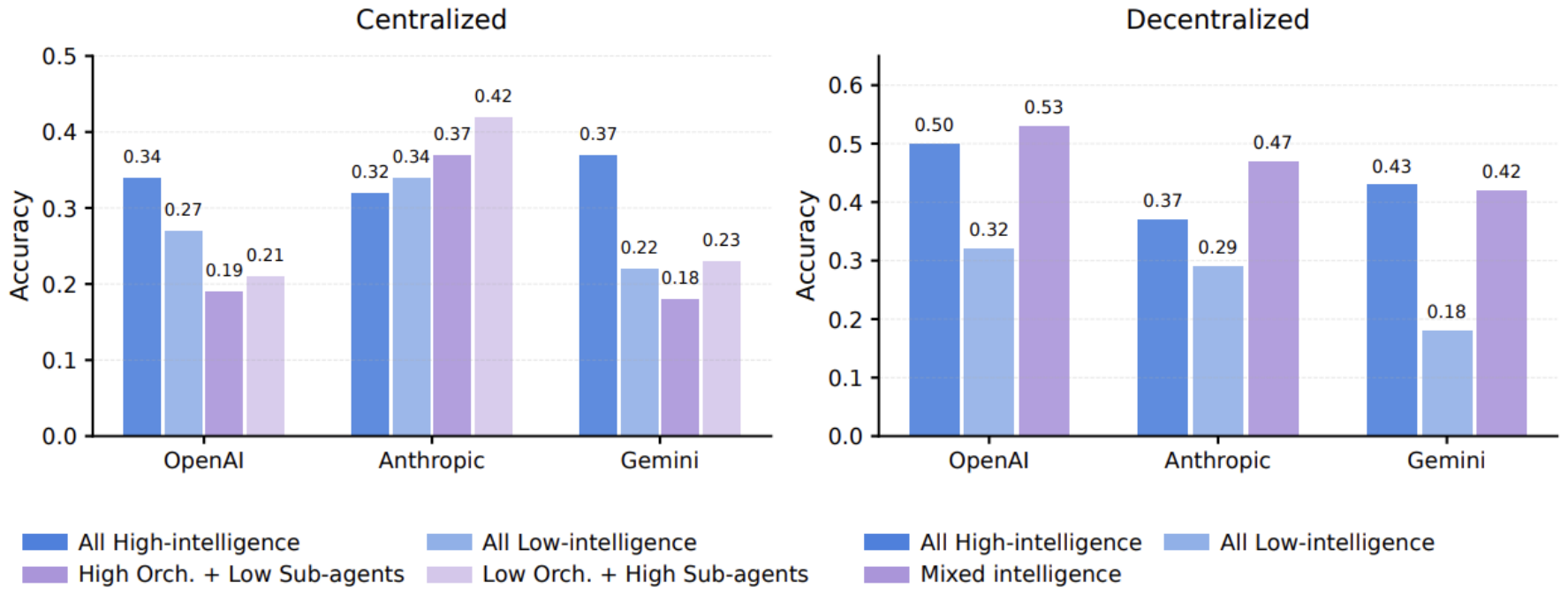
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When Does Multi-Agent Work?

Finding 4: **Mixtures of heterogeneous agents can outperform better base agents.**

We don't need expensive models everywhere. If single agent baseline > 45%, MAS hurts.

Cheap orchestrator + expensive execution sub-agents seems to work well.



[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

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When Does Multi-Agent Work?

Finding 5: **Agent-system performance can be predicted.**

A scaling law using models, tasks, MAS setup as parameters predicts the best architecture 87% of the time. From heuristics ("try a setup") to more principled science of agent design.

The complete functional form is:

$$
\begin{aligned}
P = {} & \beta_0 + \beta_1(I - \bar{I}) + \beta_2(I - \bar{I})^2 + \beta_3 \log(1 + T) + \beta_4 \log(1 + n_a) \\
& + \beta_5 \log(1 + O\%) + \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1 + A_e^{\text{trace}}) \\
& + \beta_{10} P_{\text{SA}} + \beta_{11}(I \times E_c) + \beta_{12}(A_e^{\text{trace}} \times P_{\text{SA}}) \\
& + \beta_{13}(O\% \times T) + \beta_{14}(R \times n_a) + \beta_{15}(c \times I) \\
& + \beta_{16}(E_c \times T) + \beta_{17}(P_{\text{SA}} \times \log(1 + n_a)) \\
& + \beta_{18}(I \times \log(1 + T)) + \beta_{19}(A_e^{\text{trace}} \times T) + \varepsilon,
\end{aligned}
$$

■ $p < 0.001$ ■ Significant ($p < 0.05$) ■ Non-significant ($p > 0.05$)

- Model capability (I)
- System configuration (n_a)
- Task properties (T, P_{SA})
- Coordination metrics (E_c, $O\%$, A_e, c, R)

Base model intelligence.

Num agents.

Tool count, single agent baseline.

Efficiency, overhead, trace error, message density, redundancy.

[Kim et al., Towards A Science of Scaling Agent Systems. Nature Machine Intelligence 2026]

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Multi-Agent Evolution

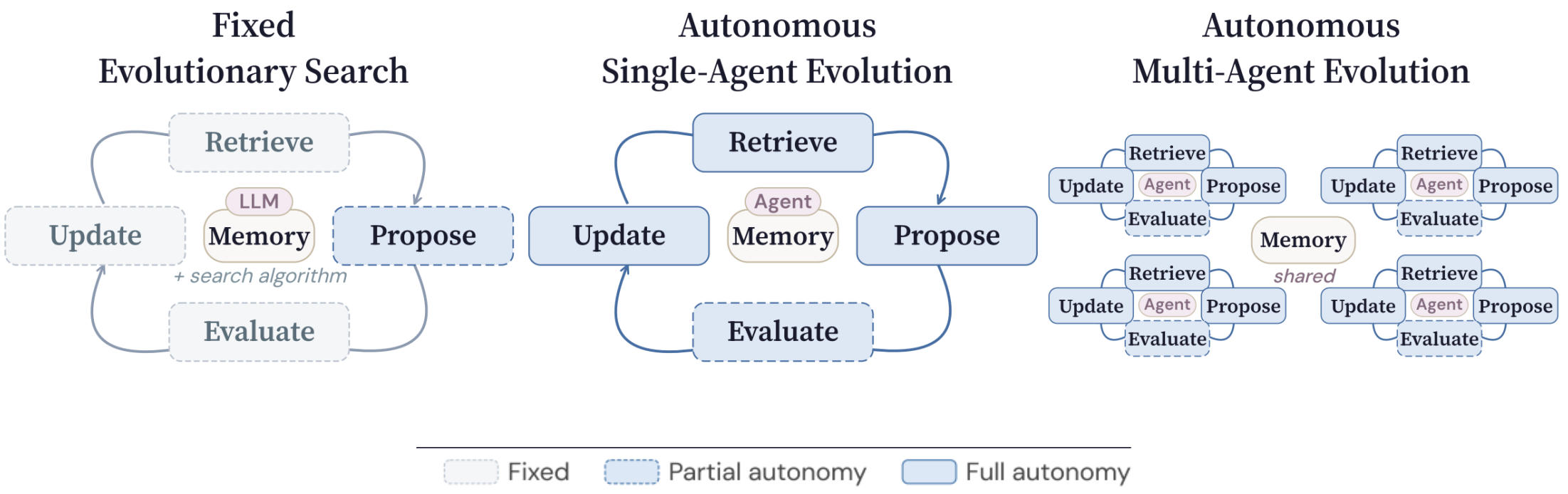
Given multi-agent systems don't always work and depend on critical hyperparameters (e.g., task, base models, topologies), there have been recent attempts at automating the multi-agent design space, instead of relying on hand engineering.

i.e., Multi-agent evolution/optimization.

What can be evolved?

- Individual models and agents (i.e., models, context, harness)
- Multi-agent system (i.e., agent roles, communication, topologies)
- Shared information (i.e., common memory, tools, skills)

CORAL for Multi-Agent Evolution



[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

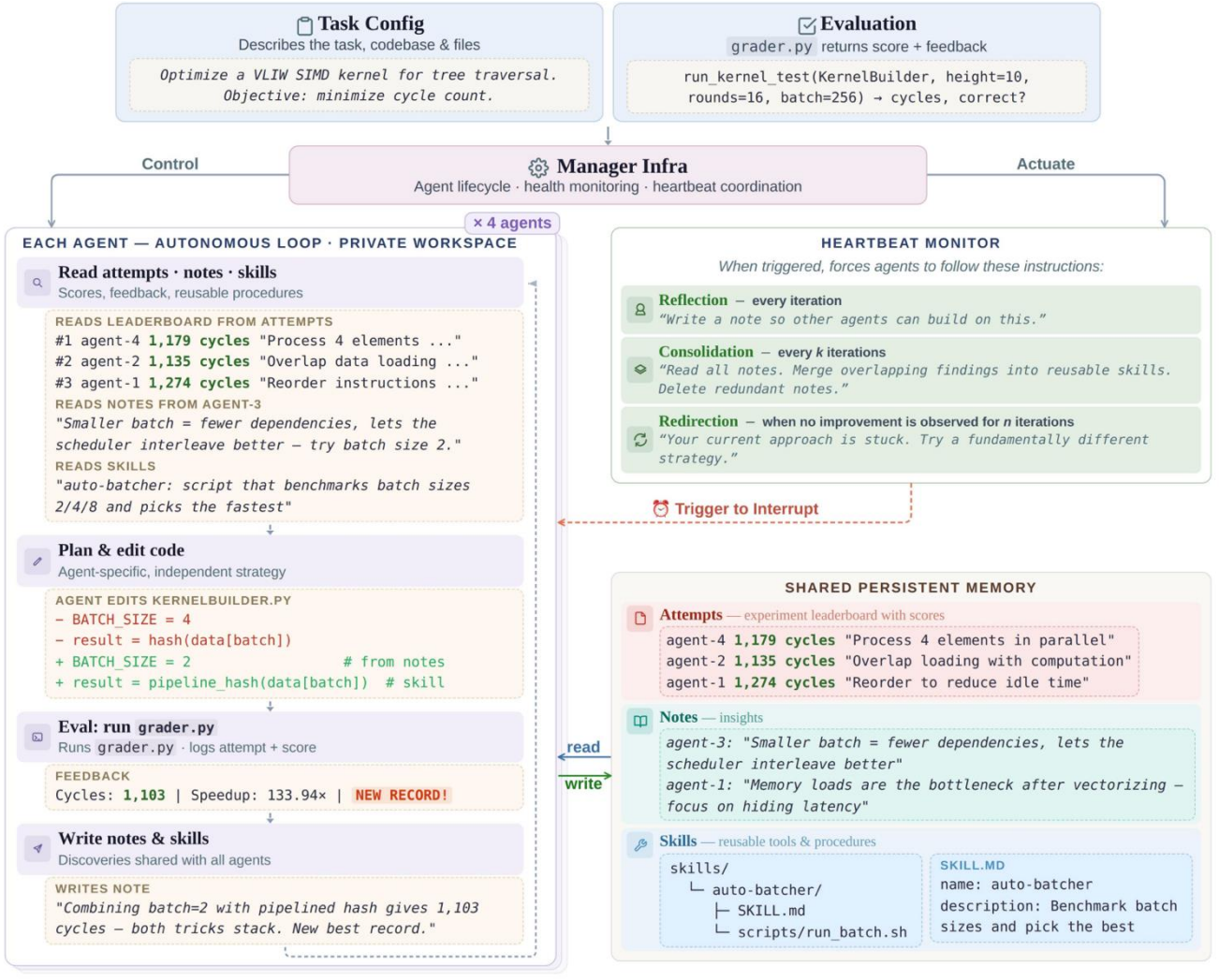
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CORAL: Multi-Agent Evolution

https://github.com/Human-Agent-Society/CORAL

Key ideas:

- **Multi-agent specialization**
- **Increased autonomy**
- **Shared memory**
- **Heartbeat mechanism**

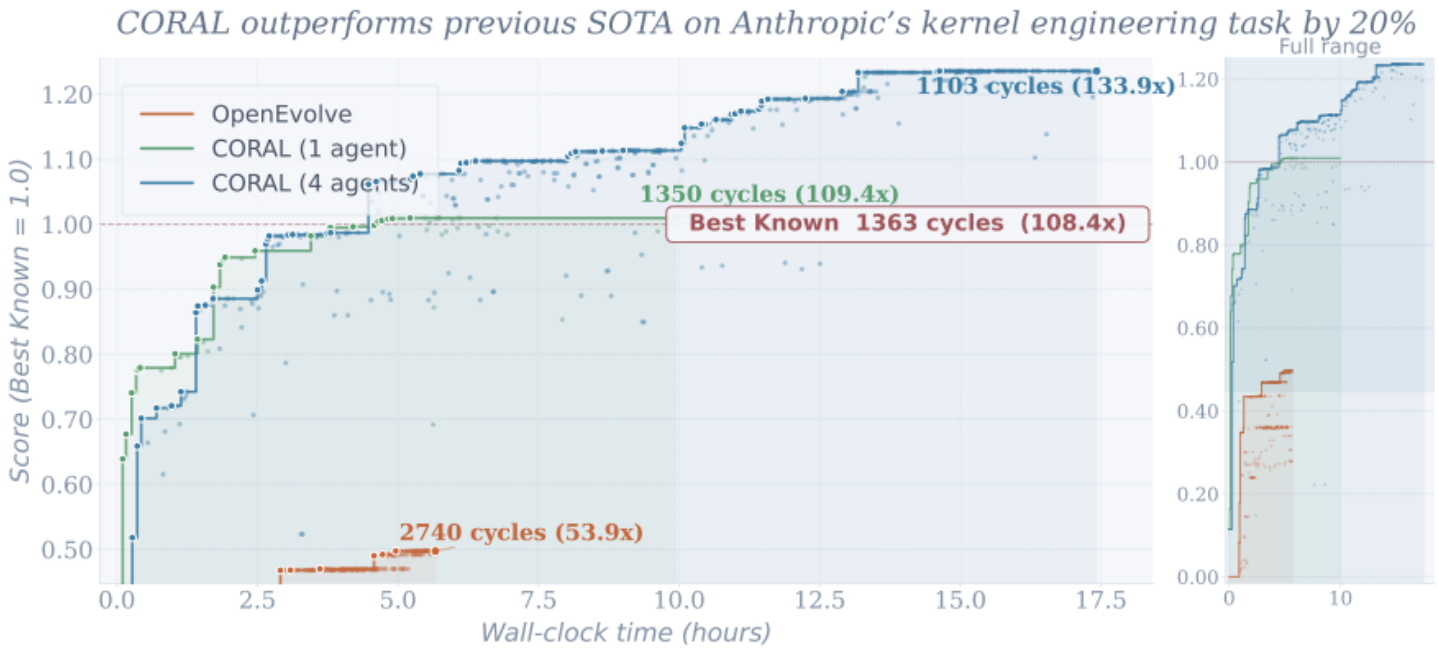


[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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Kernel Engineering

https://github.com/Human-Agent-Society/CORAL



	Model	Evals ↓	Best Cycles ↓	Speedup ↑	Imp. Rate ↑
• Best Known	Anthropic	–	1363	108.4x	–
• OpenEvolve	Opus 4.6	363	2740	53.9x	3% (12/363)
• CORAL (1 agent)	Opus 4.6	56	1350	109.4x	43% (24/56)
• CORAL (4 agents)	Opus 4.6	596	1103	133.9x	9% (54/596)

The task is to optimize a GPU kernel to minimize execution cycles from a 147,734-cycle baseline. Score = baseline / cycles, normalized so the previous best known result (1,363 cycles) equals 1.0. All methods use Claude Opus 4.6.

Operations research

→ Systems engineering

Scientific discovery

[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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More Results

https://github.com/Human-Agent-Society/CORAL

New SOTA on 8 tasks spanning math, algorithms, and system optimization

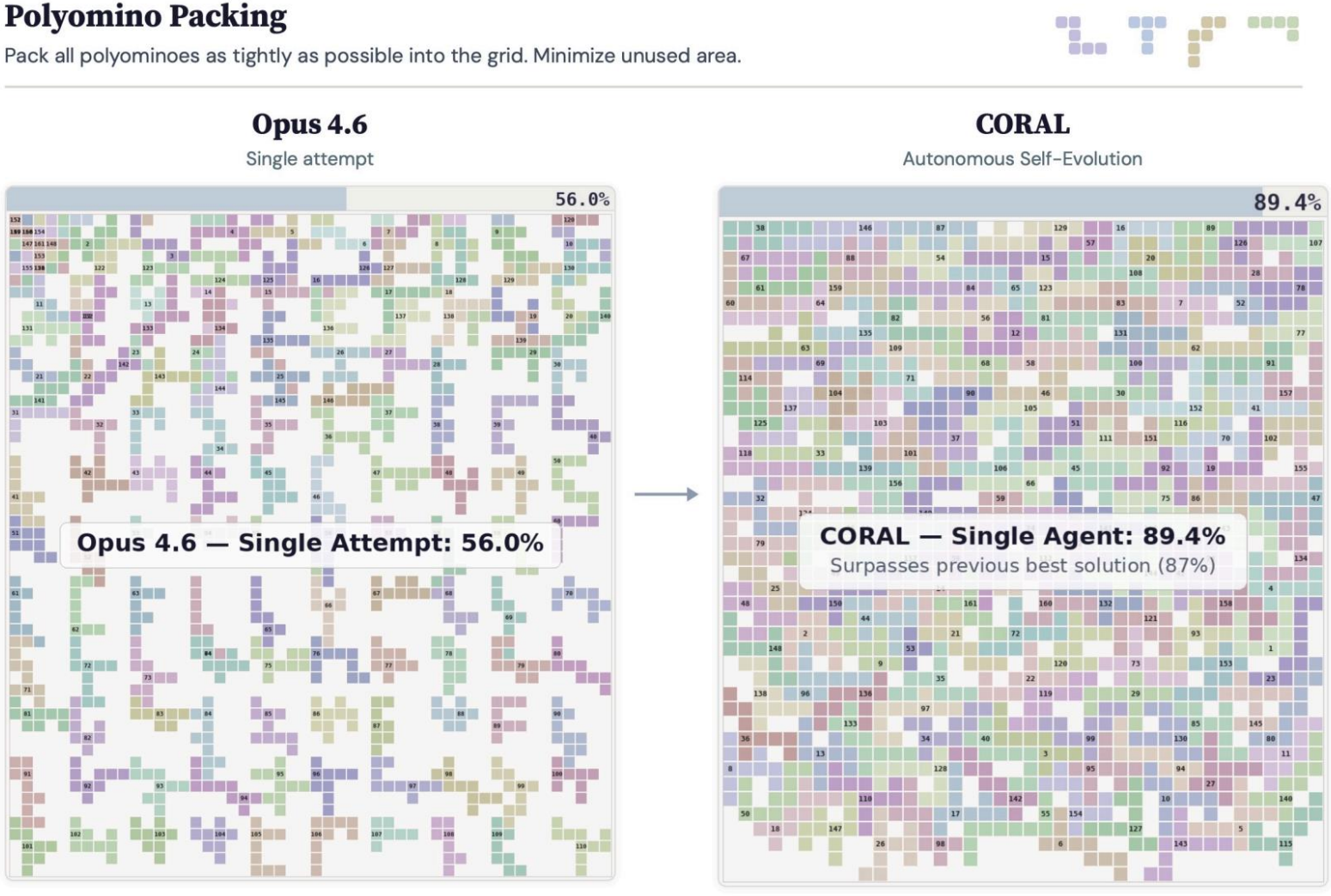
	Task	Final Score					Impr. Rate (%)				# Evals			
		SOTA	OE	SE	EvoX	CORAL	OE	SE	EvoX	CORAL	OE	SE	EvoX	CORAL
Math	Circle-Pack. ↑	2.6359	2.6293	2.6001	2.6320	**2.6360**	7.0	19.4	27.1	**100.0**	100	62	48	**11**
	Signal Proc. ↑	0.7429	0.6420	0.8171	0.7306	**0.8229**	11.8	8.6	21.1	**30.3**	85	58	71	**56**
	Erdős Over. ↓	**0.38088**	0.38188	0.38156	0.38125	0.38089	9.5	15.7	27.8	**36.8**	84	89	36	**19**
	MMD-16-2 ↓	**12.89**	12.92	**12.89**	12.96	**12.89**	8.2	9.8	33.3	**83.3**	97	82	18	**6**
	MMD-14-3 ↓	**4.16**	4.21	4.46	4.46	**4.16**	25.0	9.4	24.5	**75.0**	12	96	53	**8**
	3rd-Autocorr. ↓	**1.4557**	1.4731	1.4812	1.5552	**1.4557**	6.2	32.4	12.0	**60.0**	97	37	75	**5**
System	EPLB ↑	0.145	0.127	0.129	0.146	**0.149**	8.3	12.6	10.0	**78.9**	60	87	100	**19**
	PRISM ↑	**26.26**	**26.26**	**26.26**	**26.26**	**26.26**	31.6	18.8	25.0	**100.0**	19	32	16	**3**
	LLM-SQL ↑	0.730	0.716	0.724	0.726	**0.731**	6.7	23.8	6.0	**53.3**	100	21	83	**15**
	Txn Sched. ↑	4348	3774	3802	3984	**4566**	10.1	16.0	16.0	**27.3**	99	25	94	**22**
	Cloudcast ↓	632.7	627.2	627.8	623.5	**618.4**	7.2	20.8	12.3	**33.3**	69	24	65	**9**

[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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More Results

https://github.com/Human-Agent-Society/CORAL



[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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Multi-Agent Improves Over Single Agent

Multi-agent evolution gives gains ranging from 4% to 20%

	Runtime	Task	Final Score				Impr. Rate (%)		# Evals	
			SOTA	1-Agent	4-Agent	Gain (%)	1-Agent	4-Agent	1-Agent	4-Agent
Stress	Claude Code + Opus 4.6	Polynominoes ↑	87.0	80.2	84.2	**4.99**	42.4	19.4	33	67
		Kernel Eng. ↓	1363	1350	1103	**18.30**	43.0	9.0	56	596
Math	OpenCode + MiniMax M2.5	Circle-Pack. ↑	2.6359	2.3531	2.5391	**7.90**	20.7	28.3	29	46
		Signal Proc. ↑	0.7429	0.7174	0.7383	**2.91**	30.5	19.8	59	253
		Erdős Over. ↓	0.38088	0.39237	0.38311	**2.36**	16.7	6.9	42	58
		MMD-16-2 ↓	12.89	12.91	12.89	**0.15**	32.4	11.5	34	103
		MMD-14-3 ↓	4.16	4.53	4.19	**7.51**	63.6	15.0	11	80
		3rd-Autocorr ↓	1.4557	1.5337	1.4931	**2.65**	42.9	22.1	7	59
System	OpenCode + MiniMax M2.5	EPLB ↑	0.145	0.128	0.129	**0.78**	50.0	65.4	6	26
		PRISM ↑	26.26	25.85	26.26	**1.59**	32.6	31.7	46	82
		LLM-SQL ↑	0.730	0.693	0.730	**5.34**	83.3	71.7	6	46
		Txn Sched. ↑	4348	3704	3774	**1.89**	50.0	66.7	16	24
		Cloudcast ↓	632.7	849.4	672.8	**20.80**	72.7	66.7	11	12

[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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CORAL Analysis

https://github.com/Human-Agent-Society/CORAL

1. Importance of knowledge accumulation using memory of notes and skills.

2. Importance of multi-agent co-evolution, versus multiple independent agents.

Knowledge Accumulation (1-Agent)		
Task	**w/ Know.**	**w/o Know.**
Kernel Eng. ↓	**1350**	1601
Polyominoes ↑	**80.2**	77.3
Txn Sched. ↑	**4566**	4444

Co-evolution (4-Agent)		
Task	**Co-evol.**	**Indep. Best**
Kernel Eng. ↓	**1103**	1180
Polyominoes ↑	**84.2**	80.8
Txn Sched. ↑	**4694**	4629

[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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CORAL Analysis

https://github.com/Human-Agent-Society/CORAL

1. **Cross-agent information transfer:** On kernel engineering, 36% of attempts use another agent as their parent, these improve at 17% compared to 9% for all attempts.

2. **Exploration diversity:** Agent's vocabulary overlap by 0.3-0.4 Jaccard similarity, so more than half their explorations are unique.

3. **Contribution balance:** All agents perform roughly similar number of evolutionary attempts with similar success rates.

[Qu et al., CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery. COLM 2026]

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Lecture Summary

- Planning and task decomposition is increasingly important for efficient agents.
- Multi-agent systems bridge planning + execution, decentralized exploration, multimodal agents. But they don’t always work better!
- Multi-agent evolution can help turn hand engineering into autonomous optimization of multi-agent systems.